

Internship Report on Task 3 (Data Cleaning and Insight Generation from Survey)

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1 Introduction

This report presents an analysis of the Kaggle Data Science Survey dataset (2017–2021)[1]. The objective of this task was to clean real-world survey data, handle categorical variables, and extract meaningful insights regarding respondent demographics, professional roles, programming preferences, and salary distribution.

2 Problem / Task Details

The main objectives were:

- Perform data cleaning and remove duplicate entries.
- Handle missing and inconsistent values.
- Encode categorical variables.
- Analyze respondent behavior and professional characteristics.
- Create visual summaries of top insights.

3 Dataset Details

Key variables analyzed:

- Country (q3)
- Job Title (q5)

- Experience (q6)
- Programming Languages (q7)
- Salary Range (q25)

Top 5 Job Titles

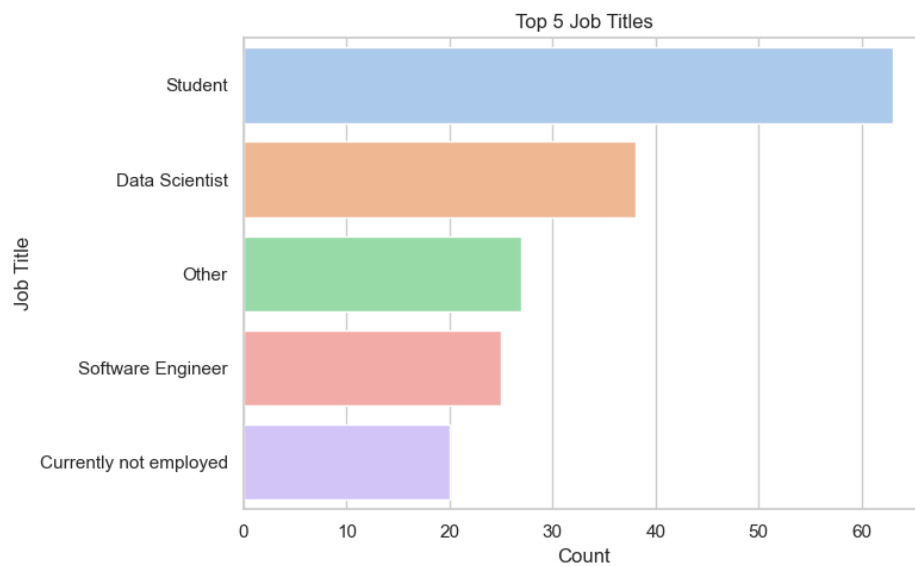


Figure 1: Top 5 Job Titles of Survey Respondents

Top 5 Countries

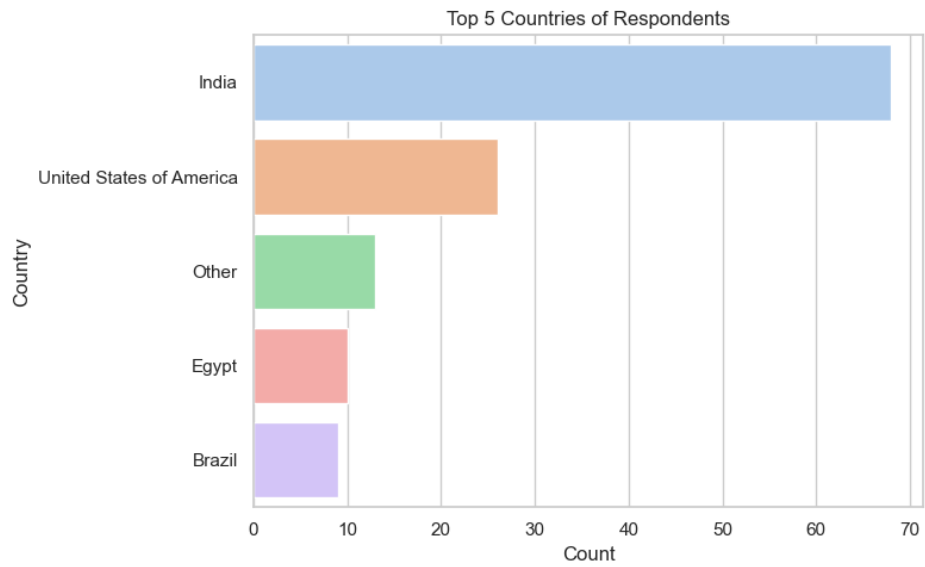


Figure 2: Top 5 Countries of Respondents

Experience Distribution

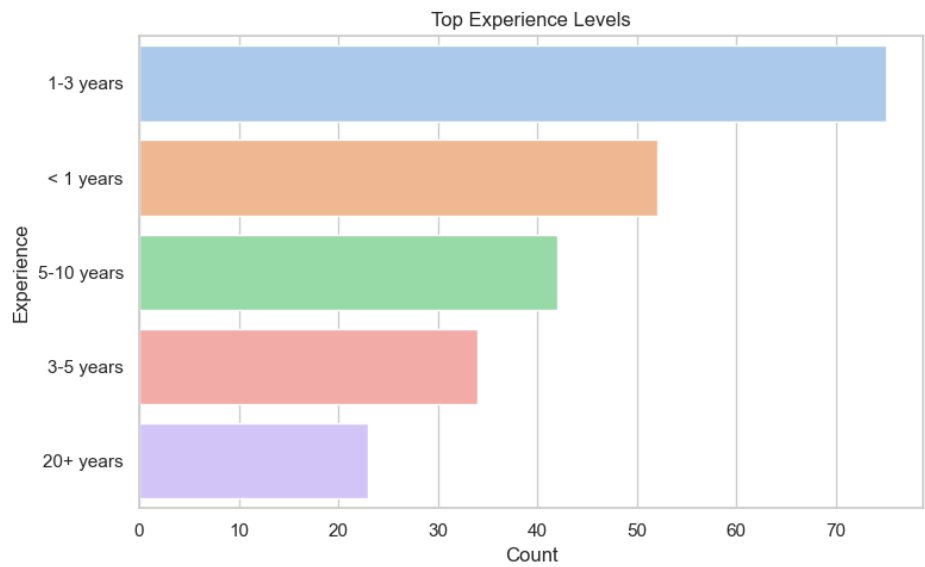


Figure 3: Experience Level Distribution

Top 5 Programming Languages

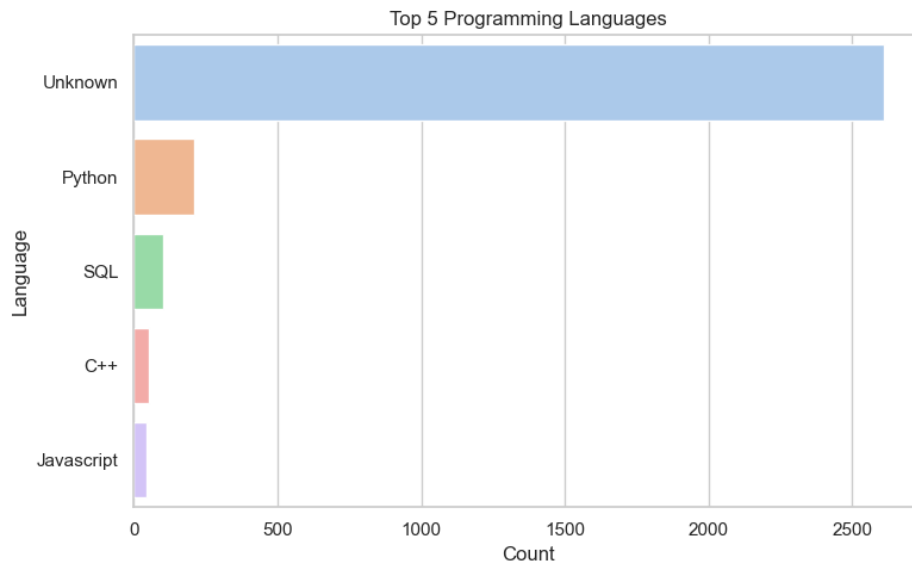


Figure 4: Top 5 Programming Languages Used

Top Salary Ranges

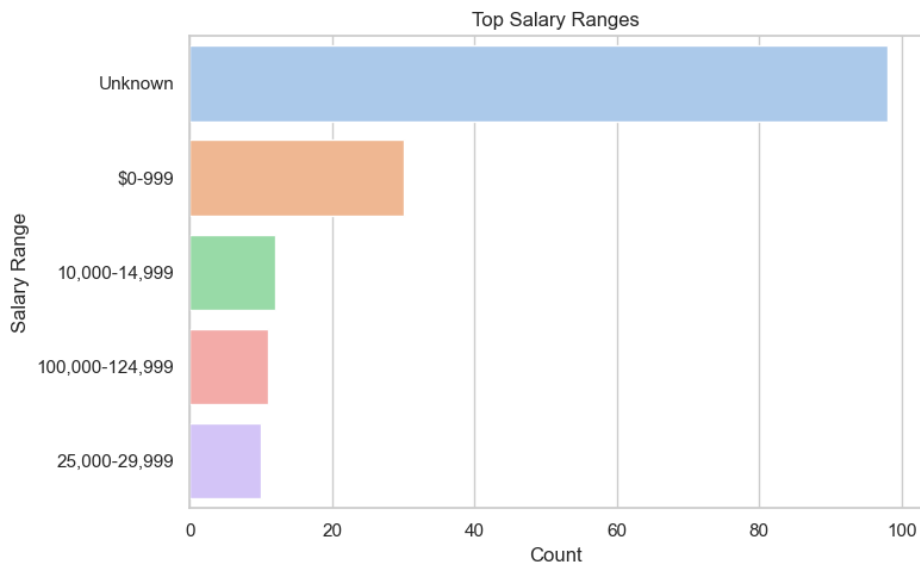


Figure 5: Top Salary Ranges of Respondents

4 Conclusion

The analysis demonstrates that Python dominates the data science ecosystem, and a significant proportion of respondents are early-career professionals or students. The survey responses are geographically concentrated in a few major countries, and salary distribution is primarily centered around mid-level income ranges.

This task highlights the importance of proper data cleaning, categorical handling, and structured exploratory data analysis when working with real-world datasets.

Bibliography

- [1] Kaggle Survey Dataset (2017–2021), Kaggle Inc. Available at: <https://www.kaggle.com>